
Backcasting for Adversarial Attacks on Time Series Forecasting

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Abstract

Adversarial attacks on time series forecasting models typically assume access to the true future values (ground truth), which is unrealistic in all real-world settings. We introduce **BATS** (Backcasting for Adversarial Attacks on Time Series), a novel black-box attack method that bypasses this limitation. BATS leverages a surrogate reverse-time model (backcaster) trained to predict past observations from current inputs, enabling gradient-based perturbations to be computed without any knowledge of future ground truth. We demonstrate the effectiveness of BATS on multiple forecasting architectures (LSTM, GRU, RNN) and compare it with standard white-box attacks, such as FGSM, black-box attacks like the boundary decision attack, and proxy-based methods using surrogate models. Our experiments indicate that BATS can significantly degrade forecasting accuracy in black-box scenarios while requiring no prior knowledge of the target model’s future outputs. This work highlights a new vulnerability in time series forecasting systems and paves the way for future research on stronger backcasting-based attacks.

1 Introduction

Time series forecasting is critical in numerous real-world applications, from financial market prediction [1] and energy demand planning to climate modeling [2], epidemic surveillance, and public health planning. In healthcare, for example, accurate forecasting of COVID-19 case counts [3] or hospital bed occupancy [4] has been critical for guiding interventions

and resource allocation. The reliability of these forecasts is crucial: small errors can have significant operational, societal, and economic consequences [5]. To model these non-linear time-dependent problems, recurrent neural networks (RNNs), specifically gated variants like GRU and LSTM, have become dominant in this domain. While suited well for the problem domain, RNNs are susceptible to adversarial perturbations [6], i.e., input modifications that can severely degrade prediction accuracy. Extensive studies in computer vision and natural language processing [7] cannot be easily adapted to time series. Unlike classification tasks with categorical outputs, forecasting produces continuous values and often follows an autoregressive structure, where each prediction depends on a sequence of past observations. This temporal dependency has the potential to amplify small perturbations over future predictions.

Adversarial attacks aim to manipulate model predictions by introducing carefully crafted perturbations to the inputs. These attacks are often categorized into two broad paradigms. In the *white-box* setting, the attacker has complete knowledge of the target model, including architecture, parameters, and sometimes access to future ground-truth values. This allows direct gradient-based methods to generate effective perturbations. In contrast, the *black-box* setting assumes the attacker has no access to model internals and may have only limited access to outputs or surrogate data. For forecasting tasks, black-box attacks face additional challenges, as the true future values required for conventional loss functions are not available at attack time, making it unclear how to generate effective perturbations.

To address these limitations, we propose **BATS** (Backcasting for Adversarial Attacks on Time Series), a novel black-box attack method that bypasses the need for future ground truth and does not require access to the outputs of the target forecasting model. Instead of relying on future values, BATS leverages a backcaster, a surrogate model trained to predict past values from the current input sequence, which provides a well-defined loss function based only on available his-

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torical data. This enables the generation of gradient-based perturbations.

The main contributions of this paper are:

- **A new black-box attack paradigm.** We introduce BATS, the first adversarial attack on time series forecasting that operates without access to model parameters, outputs, or future ground truth.
- **Backcasting-based gradient computation.** We show how reversing sequences and training a backcaster enables effective gradient-based perturbations in the absence of future values.
- **Comprehensive evaluation.** BATS is evaluated on multiple datasets and forecasting architectures (RNN, GRU, LSTM), and compared to representative white-box and black-box baselines.
- **Empirical findings.** Our experiments demonstrate that while BATS is less effective than white-box attacks, it consistently outperforms existing black-box methods under realistic constraints, thereby revealing new vulnerabilities in time series forecasting models.

2 Background

2.1 Time Series

For the purpose of this work, we consider a time series as an ordered sequence of real-valued observations $X = [x_1, x_2, \dots, x_T]$, where each x_t is a data point received from the model.

2.2 Time Series

Definition 1 (Time Series) *For the purpose of this work, we consider a time series as an ordered sequence of real-valued observations*

$$X = [x_1, x_2, \dots, x_T]$$

where T is the length and each x_t is observed at regular time intervals.

2.3 Time Series Forecasting.

Time series forecasting involves predicting future values based on historical data. Classical statistical approaches such as ARIMA [8] have been complemented by deep learning models like RNNs and their gated variants, GRUs and LSTMs, both of which can capture non-linear temporal dependencies. Given an input sequence $x_{t-L+1:t} = (x_{t-L+1}, \dots, x_t) \in \mathbb{R}^L$ of length L , a forecasting model f predicts the next value(s):

$$\hat{y}_{t+1} = f(x_{t-L+1:t}),$$

where $\hat{y}_{t+1} \in \mathbb{R}$ (univariate) or $\mathbb{R}^d$ (multivariate) is the predicted value at time $t + 1$.

2.4 Adversarial Attacks.

Adversarial attacks are small, often imperceptible perturbations added to inputs with the goal of misleading machine learning models. Although extensively studied in computer vision and classification, their adaptation to time series tasks poses unique challenges due to the temporal structure and autoregressive nature of time-based models [9]. Formally, given an input sequence x , a true target y , and a forecasting model f , the goal is to find a perturbation δ such that:

$$\arg \max_{\delta} \mathcal{L}(f(x + \delta), y) \quad \text{s.t.} \quad \|\delta\|_p \leq \epsilon$$

This optimization seeks to maximize the prediction error while keeping the perturbation within a specified norm constraint.

3 Related Work

Adversarial attacks have been widely studied across domains such as computer vision and natural language processing, where small perturbations can significantly degrade model performance [10]. The success of these methods has motivated their adaptation to time series data. In this section, we review the main lines of work and highlight the specific challenges of adversarial attacks on time series classification and forecasting.

3.1 Adversarial Attacks on Time Series Classification

Most existing research on adversarial time series focuses on classification tasks such as activity recognition or anomaly detection. Recent works have adapted gradient-based attacks such as the Fast Gradient Sign Method (FGSM) [11] or Projected Gradient Descent (PGD) [12] to time series classification, with evaluation typically based on classification accuracy. Black-box strategies such as TSAAdv [13] introduce local perturbations without requiring access to model internals, but remain restricted to classification settings. These works demonstrate the vulnerability of temporal models, yet their objectives differ from forecasting.

3.2 Adversarial Attacks on Time Series Forecasting

Contrastingly, in forecasting, the attacker's aim is to induce large deviations between predicted and true future values. Recent works have adapted gradient-based attacks such as the Fast Gradient Sign Method (FGSM) or Projected Gradient Descent (PGD) [14] to forecasting models. Concretely, given an input sequence x , the model prediction $\hat{y} = f(x)$, and the true

future target y , FGSM constructs adversarial inputs as

$$x_{\text{adv}} = x + \epsilon \text{sign}(\nabla_x L(f(x), y)),$$

where L is typically the mean squared error (MSE). These methods yield significant degradation in forecasting accuracy. However, they critically require access to the ground-truth future y at inference time, which is unrealistic in practical forecasting scenarios such as stock market prediction, weather, or energy demand.

3.3 Black-Box Attacks for Forecasting

Black-box attacks, which avoid access to model parameters, have been well studied in vision and NLP through decision-based or transfer-based approaches. Extending them to forecasting, however, is more challenging. Existing black-box strategies in other domains such as surrogate models, decision-based attacks like the Boundary Attack [15], or query-based gradient estimation [16] face unique challenges when applied to forecasting. These black-box attack methods rely either on access to the model’s outputs or on the true future target values. However, in a black-box forecasting scenario, neither are available in real-world settings: without knowing the actual future, it becomes unclear how to define a loss or compute gradients for crafting effective perturbations.

4 Methodology: Backcasting for Adversarial Attacks on Time Series

4.1 Backcasting

In backcasting, a model predicts past values based on current observations. Given an input sequence $x_{t-L+1:t} = (x_{t-L+1}, \dots, x_t)$, we reverse it to obtain $x_t^{\text{rev}} = (x_t, x_{t-1}, \dots, x_{t-L+1})$. The backcasting model f_{rev} then predicts a past value:

$$\hat{y}_{t-L} = f_{\text{rev}}(x_t^{\text{rev}}),$$

where $\hat{y}_{t-L}$ is the predicted value at time $t-L$.

4.2 Attack Pipeline

We propose a novel black-box adversarial attack method for time series forecasting, which does not require access to the ground truth. This method, named *Backcasting for Adversarial Attack on Time Series (BATS)*, leverages a surrogate model, called the *backcaster*, trained to predict past values from present inputs. The core idea is to use this reverse model to estimate gradients and craft input perturbations in the absence of future knowledge (see Figure 1 for a schematic overview).

Algorithm 1: BATS—Backcasting Adversarial Attack on Time Series

Input: Input sequence x_t , backcaster f_{rev} , perturbation ϵ

Output: Adversarial example x_t^{adv}

$x_t^{\text{rev}} \leftarrow \text{Reverse}(x_t)$;

$\hat{y}_{\text{past}} \leftarrow f_{\text{rev}}(x_t^{\text{rev}})$;

Compute loss $\mathcal{L}(\hat{y}_{\text{past}}, y_{\text{past}})$;

$g \leftarrow \nabla_{x_t^{\text{rev}}} \mathcal{L}$;

$\delta \leftarrow \epsilon \cdot \text{sign}(g)$;

$x_t^{\text{rev}'} \leftarrow x_t^{\text{rev}} + \delta$;

$x_t^{\text{adv}} \leftarrow \text{Reverse}(x_t^{\text{rev}'})$;

return x_t^{adv} ;

Iterative Extension (BATS_BIM). In addition to the one-step variant, we also implement an iterative version of BATS inspired by the Basic Iterative Method (BIM) [17]. Instead of applying a single perturbation step of size ϵ , BATS_BIM applies multiple smaller perturbations of size α (with $\alpha < \epsilon$) while projecting the adversarial example back into the valid ℓ_p -ball after each step.

4.3 Theoretical intuition

Although the forecasting loss L_f and the backcasting loss L_{rev} are distinct, their input gradients are structurally related because both models operate on a common low-dimensional feature subspace (e.g., temporal filters capturing periodicity or local dynamics). By the chain rule, we have

$$\nabla_x L_f = J_f(x)^\top r_f, \quad \nabla_x L_{\text{rev}} = P J_{\text{rev}}(x_{\text{rev}})^\top r_{\text{rev}},$$

where J_f, J_{rev} denote the Jacobians of the forecasting and backcasting models, r_f, r_{rev} are output residuals, and P is the temporal permutation operator. Here, $\nabla_x L_f$ indicates the direction in which each component of the input x should be modified to most rapidly decrease the forecasting loss, while $\nabla_x L_{\text{rev}}$ does the same for the backcasting loss, with a temporal alignment applied with P .

If we define the operator,

$$M := J_f P J_{\text{rev}}^\top,$$

then M maps the directions suggested by the backcasting gradient into the space of the forecasting output, effectively measuring modifications of x suggested by the backcast. When the models share important features, M has non-negligible singular values, meaning that some directions from the backcast are preserved and amplified rather than annihilated.

Consequently, the inner product

$$\langle \nabla_x L_f, \nabla_x L_{\text{rev}} \rangle = r_f^\top M r_{\text{rev}}$$

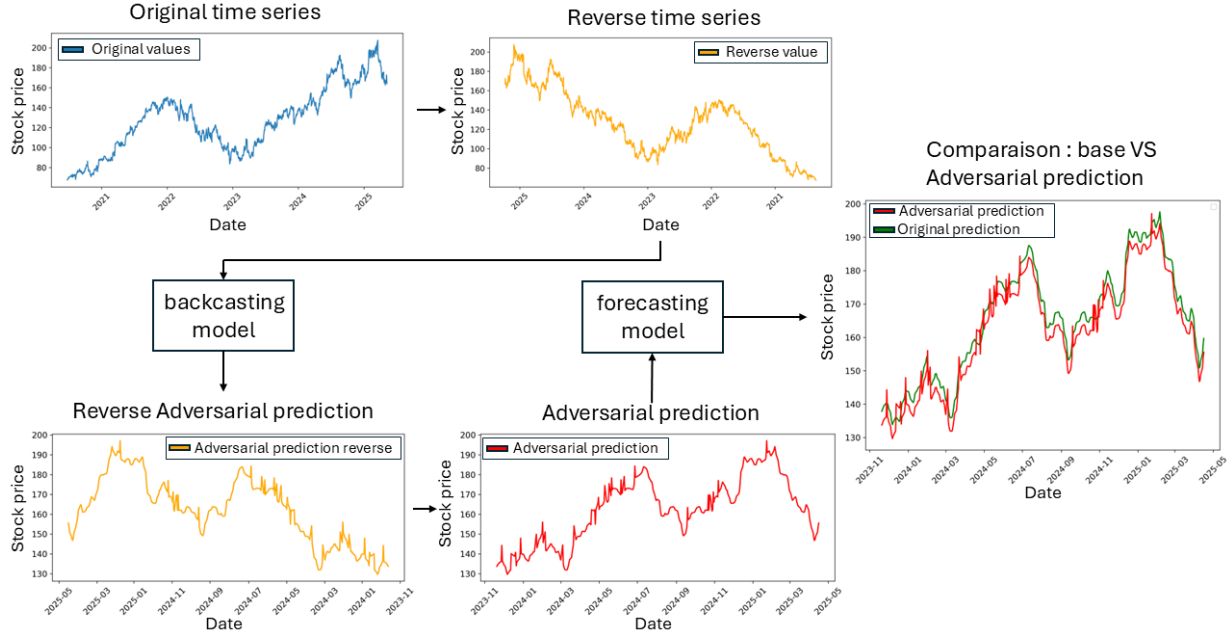


Figure 1: **Attack Pipeline of the BATS.** Based on the original time series, BATS obtains the reversed series, which is fed to a pretrained backcasting model (vanilla RNN, GRU, or LSTM). Gradients of the backcast loss are then used to craft a perturbation in reversed time. The perturbed reversed sequence is then restored to its original order to produce the adversarial example, which is fed to the separately pretrained forecasting model. The outputs of the forecaster on clean and adversarial inputs can then be compared.

quantifies the extent to which the backcasting gradient aligns with the forecasting gradient. Because M has significant singular values, this product is typically nonzero and scales with an alignment constant α and the norms of the residuals. This implies that backcasting gradients contain meaningful components along true adversarial directions of forecasting, which explains why BATS perturbations successfully transfer in practice despite relying only on reverse-time.

5 Experiments

We evaluated our approach on five different time series datasets: Google stock prices, electricity consumption, sunspots, births in the United States, and pedestrian counts. Each data set is preprocessed using min-max normalization to scale the values between 0 and 1. We apply a sliding window approach with a sequence length of 30 timesteps to generate the input sequences. The data is then split into training and testing sets using a 70/30 ratio.

BATS models are trained using batches of size 32. The target variable is always the value immediately following the input sequence. Depending on the dataset, input sequences may be univariate or multivariate. For

consistency across datasets and models, we use a simple 2-layer RNN, GRU, or LSTM with a hidden size of 64. The number of training epochs is selected empirically for each dataset to balance underfitting and overfitting, and we use the Adam optimizer with a learning rate of 0.001 and mean squared error (MSE) as the loss function. Hyperparameters were selected empirically and kept fixed across all experiments to ensure a fair comparison against several black-box and white-box baselines.

For each adversarial attack, we vary the perturbation strength ϵ across the following values: 0.01, 0.02, 0.03, 0.05, 0.07, 0.1, and 0.2. For each ϵ , we evaluate the impact of the attack on the forecasting performance using three metrics: mean absolute error (MAE), root mean squared error (RMSE), and cosine similarity between the predicted and true sequences. All reported results correspond to the average of 10 runs with different seeds to ensure robustness and reduce variance. We provide an anonymized GitHub repository containing the notebooks to reproduce the results of the presented experiments <https://anonymous.4open.science/r/backcasting-for-adversarial-attacks-on-time-series-F800/README.md>.

6 Results and Discussion

Table 1 reports the average performance of each attack method across the three evaluated architectures (vanilla RNN, GRU, and LSTM), averaged over the five datasets. The metrics reported are mean absolute error (MAE), root mean squared error (RMSE), and cosine similarity (SIM). For MAE and RMSE, higher values indicate a more successful attack (larger prediction error). For SIM, lower values indicate stronger degradation in the temporal structure of the predictions. Figure 2 complements these results by illustrating the impact of adversarial attacks on three of the five datasets (Sunspots, Electricity, Birth) across GRU, LSTM, and RNN models.

BATS effective across architectures. Across the RNN, LSTM, and GRU models, BATS consistently induces substantial performance degradation. For example, in the RNN setup with $\epsilon = 0.10$, BATS increases the MAE by approximately 92 % (from 91.86 to 176.44), while the cosine similarity (SIM) decreases by 18 % (from 0.89 to 0.73), indicating a notable change in prediction patterns. Similar trends are observed for LSTM, where MAE increases by about 80 % (from 94.15 to 169.74) and SIM decreases by 15%, and for GRU, with an increase of roughly 131 % in MAE (from 70.11 to 162.17) accompanied by a decrease of 25 %.

Discussion. Overall, the results confirm that BATS and its iterative variant (BATS_BIM) attacks significantly degrade forecasting performance in realistic black-box scenarios without requiring any access to future ground truth. While white-box FGSM yields the largest absolute errors, this comes at the cost of an unrealistic assumption in many real-world forecasting scenarios (access to future targets). Furthermore, BATS most of the time outperforms the boundary attack across different architectures and perturbation magnitudes, indicating its effectiveness as a black-box adversarial attack for time series forecasting.

BATS presents a practical middle ground. Among the baselines, *FGSM* (white-box) produces the highest MAE and RMSE increases, but it requires access to model gradients and ground truth. The transfer variant, *FGSM surrogate*, can also yield strong errors when the surrogate closely matches the target, yet this setting is unrealistic since future labels are unavailable to the attacker. In contrast, the decision-based *Boundary Attack* achieves meaningful error increases but usually underperforms BATS, leading to lower degradation across most scenarios. For instance, for the RNN model at $\epsilon = 0.20$, Boundary Attack increases the MAE by about 186 %, compared to 193 % for BATS, but only at the cost of many more

model queries. Overall, these results position BATS as a practical middle ground: more effective than decision- or transfer-based black-box methods under realistic constraints, yet less effective than fully informed white-box attacks.

Adversarial vulnerability is dataset dependent. The impact of adversarial perturbations is not uniform across datasets (cf. Figure 2), but strongly depends on the statistical properties of the time series. On the Sunspots dataset—which exhibits clear periodic cycles—BATS causes a sharp degradation in cosine similarity (from 0.70 to below 0.25 as ϵ increases). This indicates that the attack is particularly effective at disrupting the temporal structure of highly regular signals. While such disruption increases forecasting error and maximizes the attack objective, it may simultaneously make the attack more detectable by defenses monitoring temporal structure.

In contrast, on the Birth dataset, which exhibits less periodic patterns, similarity remains close to 0.99 even at high perturbation levels. On such datasets, adversarial perturbations—including BATS—tend to primarily alter magnitudes without significantly affecting structural consistency. The attack is therefore less obvious in terms of temporal shape but still damaging, as MAE and RMSE increase substantially.

Finally, for the electricity dataset, we focused on a single univariate consumption series. This series exhibits moderate fluctuations with periods of relative stability. On this data, adversarial perturbations consistently increase MAE and RMSE more than the boundary attack, but the cosine similarity decreases more gradually. This indicates that series with moderate stability, adversarial attacks primarily disrupt magnitudes rather than fully altering temporal patterns.

These observations indicate that adversarial vulnerability is dataset-dependent: structured periodic signals are most fragile in terms of shape, stable series amplify magnitude errors, and series with fluctuations but moderate long-term stability are especially exposed to a combination of both types of distortions.

Limitations While BATS consistently outperforms existing black-box methods, a few limitations remain. First, we restricted our experiments to recurrent models (RNN, GRU, LSTM); extending the analysis to more recent architectures, such as autoregressive transformers or temporal convolutional networks, would provide a broader view. Moreover, we did not explore defenses or detection mechanisms, which would be an interesting direction for future work in assessing the robustness of deployed forecasting systems against backcasting-based attacks.

Table 1: Adversarial attack results for RNN, GRU and LSTM models

Model	Attack	Metrics	ϵ						
			0 (No attack)	0.01	0.03	0.05	0.07	0.1	0.20
RNN	BATS	MAE	91.86	101.19	128.49	144.98	159.44	176.44	268.54
		RMSE	119.36	128.21	156.02	173.31	189.62	208.21	312.10
		SIM	0.89	0.84	0.69	0.69	0.73	0.73	0.81
	BATS BIM	MAE		101.19	128.37	144.43	155.35	175.11	265.07
		RMSE		128.21	155.89	172.97	185.88	208.67	312.23
		SIM		0.84	0.69	0.69	0.73	0.73	0.81
	Boundary Attack	MAE		100.92	121.10	136.53	154.31	177.70	263.09
		RMSE		130.55	160.17	182.86	208.64	247.23	365.25
		SIM		0.84	0.78	0.76	0.75	0.75	0.74
	FGSM Surrogate	MAE		130.68	191.52	241.32	303.21	366.10	615.56
		RMSE		159.54	225.60	281.06	351.37	423.15	704.64
		SIM		0.81	0.75	0.75	0.75	0.74	0.72
	FGSM	MAE		150.64	212.13	264.53	331.44	398.50	652.80
		RMSE		176.98	244.47	304.54	381.59	458.53	746.10
		SIM		0.77	0.72	0.72	0.72	0.72	0.70
LSTM	BATS	MAE	94.15	99.13	114.47	129.63	150.48	169.74	251.82
		RMSE	122.18	126.25	142.44	161.26	188.07	212.93	316.17
		SIM	0.91	0.87	0.81	0.79	0.77	0.78	0.77
	BATS BIM	MAE		98.69	112.42	126.26	140.91	162.11	233.86
		RMSE		125.50	139.89	156.21	173.52	198.79	280.65
		SIM		0.88	0.83	0.82	0.82	0.81	0.80
	Boundary Attack	MAE		101.69	118.39	131.38	148.95	169.39	242.80
		RMSE		131.19	157.41	177.75	204.92	235.21	338.26
		SIM		0.86	0.79	0.76	0.75	0.75	0.74
	FGSM Surrogate	MAE		131.01	186.22	241.06	308.01	378.16	618.69
		RMSE		160.67	220.73	281.36	356.04	435.49	705.54
		SIM		0.81	0.76	0.75	0.74	0.72	0.71
	FGSM	MAE		143.39	211.95	275.30	354.53	431.58	695.11
		RMSE		171.60	243.25	312.75	399.89	484.46	773.24
		SIM		0.81	0.75	0.74	0.74	0.73	0.71
GRU	BATS	MAE	70.11	97.53	108.81	121.02	140.02	162.17	266.01
		RMSE	100.08	124.62	135.30	148.29	168.61	192.38	307.10
		SIM	0.91	0.84	0.71	0.68	0.68	0.68	0.67
	BATS BIM	MAE		75.14	86.69	97.27	108.76	127.78	210.92
		RMSE		103.30	113.79	124.10	136.02	156.11	245.48
		SIM		0.89	0.77	0.75	0.75	0.76	0.76
	Boundary Attack	MAE		78.47	97.22	113.83	135.32	152.95	237.93
		RMSE		110.40	137.19	164.03	194.04	219.54	335.34
		SIM		0.86	0.79	0.77	0.76	0.75	0.74
	FGSM Surrogate	MAE		105.00	160.95	213.61	283.97	356.43	597.65
		RMSE		138.61	199.98	258.33	337.54	419.18	689.42
		SIM		0.83	0.77	0.76	0.75	0.75	0.72
	FGSM	MAE		123.13	191.45	253.94	335.63	416.14	665.23
		RMSE		151.06	221.88	290.22	381.08	470.48	749.73
		SIM		0.81	0.75	0.75	0.74	0.74	0.72

Note: Bold values denote the black-box attack (BATS, BATS.BIM, Boundary Attack) that produced the highest MAE for a given model and ϵ . FGSM and FGSM Surrogate are not emphasized.

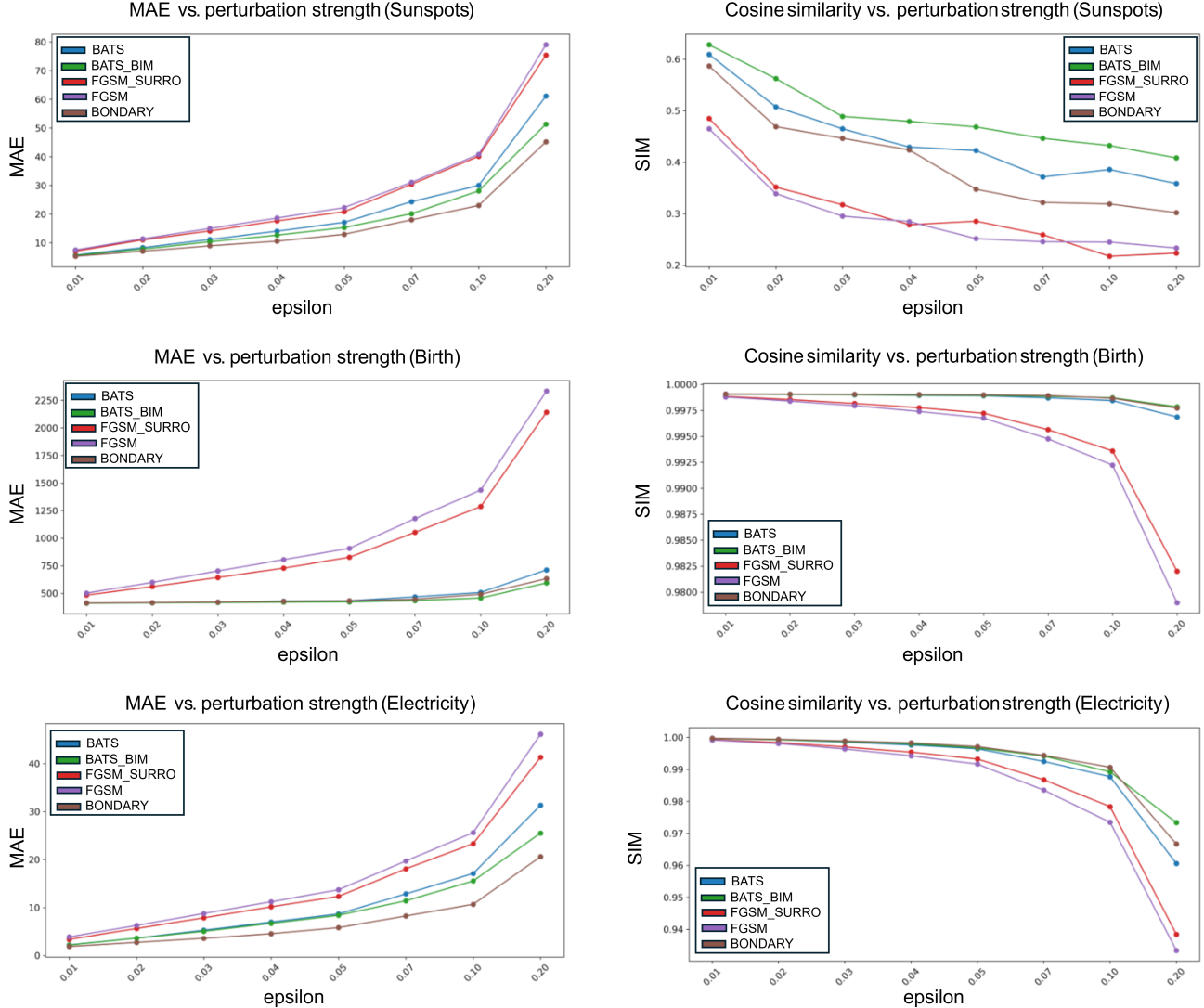


Figure 2: MAE and cosine similarity vs. perturbation strength (ϵ) across three representative datasets. The impact of adversarial perturbations varies depending on dataset structure: periodic series (Sunspots) are most fragile in terms of temporal shape, while less regular datasets (Birth) primarily amplify magnitude errors.

7 Conclusion

In this paper, we introduced BATS, a novel black-box adversarial attack framework for time series forecasting that bypasses the unrealistic assumption of access to future ground truth. By leveraging a surrogate backcasting model trained to predict past observations from the current input sequence, BATS enables effective gradient-based perturbations without any knowledge of the target model’s outputs or parameters.

Our extensive experiments across multiple datasets and recurrent architectures demonstrate that BATS consistently degrades forecasting accuracy, outperforming existing black-box baselines while remaining more practical than fully informed white-box attacks.

We further observed that the vulnerability of forecasting models is dataset-dependent: structured periodic signals are particularly fragile in terms of temporal shape, while more irregular series primarily amplify magnitude errors. These insights suggest that adversarial robustness cannot be evaluated in isolation but must account for the statistical properties of the underlying time series. This reveals an overlooked vulnerability of forecasting systems that are increasingly deployed in high-stakes domains such as finance, energy, and healthcare, where even moderate prediction errors may lead to severe consequences.

While our study focused on recurrent models, extending the analysis to modern architectures such as Transformers or temporal convolutional networks is a nat-

ural next step. Another promising direction lies in developing defense mechanisms against backcasting-based attacks, including adversarial training strategies or perturbation detection methods.

We hope that this work not only underscores the risks associated with forecasting under adversarial settings but also inspires the design of more robust and trustworthy time series forecasting systems capable of withstanding such black-box threats.

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Checklist

- For all models and algorithms presented, check if you include:
 - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] See Sections 3–4 (Methodology and Algorithm).

- (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Not Applicable]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes]
- 2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Not Applicable]
 - (c) Clear explanations of any assumptions. [Yes]
- 3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes] See the project README for detailed reproduction instructions
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] All experiments were conducted on Google Colab using free GPUs
- 4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [Yes]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
- 5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]